

QPE (3 ancilla, 2 target)

$$|\psi_0\rangle = |0\rangle_{\text{anc2}} \otimes |0\rangle_{\text{anc1}} \otimes |0\rangle_{\text{anc0}} \otimes \frac{|0\rangle \otimes |0\rangle}{\text{target}(2)}$$

Grouped ancilla & target

$$|\psi_0\rangle = |000\rangle \otimes |00\rangle$$

As a vector : 5 qubits $\rightarrow 2^5 = 32$ numbers

Apply X gate (Pauli X gate on target)

$$X|00\rangle = |11\rangle$$

Now full state equation become,

$$|\psi_1\rangle = |000\rangle \otimes |11\rangle$$

Apply H gate on ancillas

$$|\psi_2\rangle = \left[\frac{(|0\rangle + |1\rangle)}{\sqrt{2}} \otimes \frac{(|0\rangle + |1\rangle)}{\sqrt{2}} \otimes \frac{(|0\rangle + |1\rangle)}{\sqrt{2}} \right]$$

Now full state equation become,

$$|\psi_2\rangle = \frac{1}{\sqrt{2^3}} (|0\rangle + |1\rangle) \otimes |11\rangle$$

or,

$$|\psi_2\rangle = \frac{1}{\sqrt{8}} \sum_{x=0}^7 |\psi\rangle \otimes |11\rangle$$

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Apply controlled U gate on target
ancillas

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$$|\psi_3\rangle = \left(\frac{1}{\sqrt{8}}\right) \sum_{x=0}^7 e^{i \cdot x \cdot 0} |x\rangle \otimes |11\rangle$$

Now, the same equation also the full state equation is

$$|\psi_3\rangle = \left[\frac{|0\rangle + e^{i \cdot \underset{\text{ancilla 1}}{1} \cdot 0} |1\rangle}{\sqrt{2}} \right] \otimes \left[\frac{|0\rangle + e^{i \cdot \underset{\text{ancilla 2}}{2} \cdot 0} |1\rangle}{\sqrt{2}} \right] \otimes$$

Ancilla 1

$$\left[\frac{|0\rangle + e^{i \cdot \underset{\text{ancilla 2}}{4} \cdot 0} |1\rangle}{\sqrt{2}} \right]$$

ancilla 2

Replace x with $2^0, 2^1, 2^2$ which is 1, 2, 4

now, full state equation is

$$|\psi_3\rangle = \left(\frac{|0\rangle + e^{i0} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + e^{i20} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + e^{i40} |1\rangle}{\sqrt{2}} \right)$$

Ancillas Target

This is final equation of the complete state after applying controlled Ugate on target.

Note, Before we jump into inverse QFT let's understand what is 'x'?

x is one combination of all 3 ancillas together,

x	3 bits	ancilla2	ancilla1	ancilla0
0	000	0	0	0
1	001	0	0	1
2	010	0	1	0
3	011	0	1	1
4	100	1	0	0
5	101	1	0	1
6	110	1	1	0
7	111	1	1	1

After H gate on ancillas, the 3 ancillas are in Superposition of all 8 combinations at once, so, the state holds every x from 0 to 7 together, each weight $1/8$

The phase $e^{ix \cdot \theta}$ also comes from the digits: each ancilla that is 1 adds its own turn.

Eg. $x=5=101$

$$\begin{array}{ccc} a_2 & a_1 & a_0 \\ \downarrow & \downarrow & \downarrow \\ 2^2 & 2^1 & 2^0 \end{array}$$

ancilla 2^2 adds $40 \times 1 = 40$

ancilla 2^1 adds $20 \times 0 = 0$

ancilla 2^0 adds $10 \times 1 = 10$

Sum, $40 + 0 + 10 = 50 = x \cdot \theta$

Inverse Quantum Fourier Transform

A quantum operation on n qubits that maps each basis state $|x\rangle$ to an equal superposition of all basis states $|k\rangle$, with that "turn back" by an amount proportional to $x \cdot k$.

$$\text{IQFT} |x\rangle = \left(\frac{1}{\sqrt{N}}\right) \sum e^{-2\pi i \cdot x \cdot k / N} |k\rangle$$

Here, Applying Inverse QFT on QPE

In QPE, it is the step that converts the phases $e^{ix \cdot \theta}$ stored in ancillas into the bitstring k , so that

$\theta \approx 2\pi k/N$ can be read by measuring

Here we are going to introduce new term x'
 x' = labels of results after the IQFT, where the information is now in which single label

Pseudo idea

$$\text{amplitude of } x' = \frac{1}{\sqrt{8}} \left[(\text{term for } x=0) + (x=1) + \dots + x=7 \right]$$

Example

$$\text{Term for } x = \underbrace{e^{ix \cdot \theta}}_{\text{from } U} \times \underbrace{e^{-ix \cdot 2\pi \frac{x'}{\sqrt{8}}}}_{\text{from IQFT}}$$

Write out for 2 output ($\theta = 90^\circ$)

$$x' = 1; \frac{1}{\sqrt{8}} \times \frac{1}{\sqrt{8}} \left[e^{i0.45} + e^{i.145} + e^{i.2.45} + \dots + e^{i.7.45} \right] = 0$$

$$x' = 2; \frac{1}{8} \left[e^{i0.0^\circ} + e^{i1.0^\circ} + \dots + e^{i7.0^\circ} \right] = \frac{1}{8} \times 8 = 1$$

Confuse why x'_1 has 45° and $x'=2$ has 0°

Here in above example $\theta = 90^\circ$ in both:

leftover are 45° and 0°

confuse? Don't worry!!

$$\begin{aligned} x' = 1: & 90^\circ - 1 \times 45^\circ \\ & 90^\circ - 45^\circ = 45^\circ \end{aligned}$$

$$\begin{aligned} x' = 2: & 90^\circ - 2 \times 45^\circ \\ & 90^\circ - 90^\circ = 0 \end{aligned}$$

$$x'=2: 90^\circ - 2 \times 45^\circ$$

$$90^\circ - 90^\circ = 0$$

Why 45° is fixed?

→ This is depend on how many ancilla's we have

$$\text{Ancillas} = 3$$

$$2^3 = 8$$

$$\text{Total angle} = \frac{360}{8} = 45^\circ$$

let's try for $x'=3 = 90^\circ - 3 \times 45^\circ \Rightarrow 90 - 135 = -45^\circ$

$$x'=3 = \frac{1}{8} (e^{i0(-45)^\circ} + e^{i1(-45)^\circ} + \dots + e^{i7(-45)^\circ}) = 0$$

$$|\psi_4\rangle = \frac{1}{8} \sum_{x'=0}^7 e^{ix(\theta - x'45^\circ)} |x'\rangle \otimes |11\rangle$$

Here leftover = $\theta - x'45^\circ$ [$x' = 0$ to 7]

final state equation;

$$|\psi_4\rangle = \frac{1}{8} \sum_{x'=0}^7 e^{ix(\theta - x'45^\circ)} |x'\rangle \otimes |11\rangle$$

Note, x' goes from 0 to 7, which is 8 possible values and each one is written with 3 binary digits [$2^3 = 8$]

x'	0	1	2	3	4	5	6	7
Binary	000	001	010	011	100	101	110	111

Suppose, $\theta = 90^\circ$, leftover per step is $90^\circ - x' \cdot 45$

x'	leftover	$x=0$	$x=1$	$x=2$	$x=3$	$x=4$	$x=5$	$x=6$	$x=7$	Sum /8
0	90	1	i	-1	$-i$	1	i	-1	$-i$	0
1	45	1	$a+ai$	i	$-a+ai$	-1	$-a-ai$	$-i$	$a-ai$	0
2	0	1	1	1	1	1	1	1	1	1
3	-45	1	$a-ai$	$-i$	$-a-ai$	-1	$-a+ai$	i	$a+ai$	0
4	-90	1	$-i$	-1	i	1	$-i$	-1	i	0
5	-135	1	$-a-ai$	i	$a-ai$	-1	$a+ai$	$-i$	$-a+ai$	0
6	-180	1	-1	1	-1	1	-1	1	-1	0
7	-225	1	$-a+ai$	$-i$	$a+ai$	-1	$a-ai$	i	$-a-ai$	0

Are you confuse what is $a+ai$?

wait, it's simple

$$\begin{aligned} e^{i45^\circ} &= \cos 45^\circ + i \sin 45^\circ \\ &= 0.707 + i 0.707 \\ &= a+ai \quad \checkmark \end{aligned}$$

In above IQFT only $x'=2$ has
amplitude $\pm$ so $\theta=90^\circ$

Note

On paper, we have to plug θ in to
predict what circuit will output.

A real quantum computer doesn't
do this: θ is hidden inside U , the
gate physically do the turning
and cancelling and you only
measured x' . so in the real
QPE you don't know θ , you
read it from x' .

Does the result always match θ ?

No, only when θ lands exactly on a slice

θ	Is it on slice?	What you Measure	Estimate
90°	Yes ($2 \times 45^\circ$)	010 every time	90° exact
27°	NO (bet ⁿ 0-45)	001 $\approx 58\%$ 000 $\approx 26\%$	45° most often, off by 18°

So the result is always the nearest slice (most likely), not always the exact θ .

Noise add

Symbol	Meaning	Our example
a	number of ancillas qubits	3
N	no. of possible answers = 2^a	8
θ	the hidden angle of U	90°
X	8 ancilla pieces before IQFT	01...7
x'	the ans after IQFT (0 to 7)	$x' = 2 = 010$
$A(x')$	Amplitude of answer x' $A(x') = \left(\frac{1}{\sqrt{2^a}}\right) \times (\text{sum of 8 arrows})$	For $x' = 2$ Ans = 1
$P(x')$	Probability of answer $x' = A(x')^2$	For $x' = 2$ Ans = 1
P	error chance of one gate	0.03
S	Survival: Chance that no gate makes an error	0.885
q	chance of misreading one bit	0.02